# CPE 486/586 Classwork 02: Linear Algebra

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 02, 2026 (In-Class)

1. Compute the dot product $\mathbf{v} \cdot \mathbf{w}$ of two vectors $\mathbf{v} = [6, 7, 9]$ and $\mathbf{w} = [12, 0, -3]$. **(5 Points)**

**Answer**

2. Compute L2 norm of vector $\mathbf{w} = [12, 0, -3]$. **(5 Points)**

**Answer**

3. Consider a code snippet below:

```
import numpy as np
v = np.array([[1, 2, 3],[8, 0, 3]])
```

How many columns and rows vector $v$ has? **(2 Points)**.

```
import numpy as np
v = np.array([[1],[2],[3],[4]])
print(v.shape)
```

What is the output of the `print` statement in the above code-snippet? **(3 Points)**

**Answer**

4. Consider a code-snippet:

```
import numpy as np
v = np.array([8, 0, 3])
norm_v = np.linalg.norm(v, ord=1)
print(norm_v)
```

What is the output of the `print` statement in the above code-snippet? **(5 Points)**

**Answer**